This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options. ***1 point***

- [ ] If a 3×3 matrix A is similar to the identity matrix of order 3, then A must be the identity matrix of order 3.
- [ ] If a 3×3 matrix A is similar to a diagonal matrix of order 3, then A must be a diagonal matrix of order 3.
- [ ] If A and B are similar matrices, then they are also equivalent matrices.
- [ ] If A and B are equivalent matrices, then they are also similar matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If a 3×3 matrix A is similar to the identity matrix of order 3, then A must be the identity matrix of order 3.
If A and B are similar matrices, then they are also equivalent matrices.

Consider the linear transformation given below:
$T : \mathbb{R}^3 \to \mathbb{R}^2$

$T(x, y, z) = (x - y, y + z)$

Consider two ordered bases $\beta_1 = \{(1,0,0), (0,1,0), (0,0,1)\}$, and $\beta_2 = \{(1,0,1), (0,1,1), (1,1,0)\}$ of $\mathbb{R}^3$ and consider the standard ordered basis $\gamma = \{(1,0), (0,1)\}$ of $\mathbb{R}^2$. Let A and B be the matrices corresponding to the linear transformation T with respect to the bases β_1 and β_2 for the domain, respectively and γ for the co-domain. Let Q and P be matrices such that $B = QAP$.

Answer the questions 2,3, and 4 using the above information.

2) Which of the following matrices represents A? **1 point**

- $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

3) Which of the following matrices can be P? **1 point**

- There does not exist any matrix P which satisfies the property $B = QAP$.

- [ ] $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$
- [ ] $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$
- [ ] $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

4) Which of the following statements is true for Q? **1 point**

- [ ] Q can be the identity matrix of order 3.
- [ ] Q can be the identity matrix of order 2.
- [ ] Whatever the matrix P we choose, Q is always unique.
- [ ] Q can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Q can be the identity matrix of order 2.
Q can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

5) Choose the set of correct options. ***1 point***

- [ ] If two square matrices of the same order have the same determinants, then they must be similar to each other.
- [ ] Similar matrices have the same rank.
- [ ] If A and B are similar matrices, and $Ax = b$ has a unique solution, then $Bx = b$ also has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Similar matrices have the same rank.
If A and B are similar matrices, and $Ax = b$ has a unique solution, then $Bx = b$ also has a unique solution.

6) Choose the set of correct options. **1 point**

- [ ] If det$A$ or det$B$ is nonzero, then AB and BA are similar.
- [ ] $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ are similar.
- [ ] $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 11 \\ 0 & 4 \end{bmatrix}$ are similar.
- [ ] $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are not similar.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If det$A$ or det$B$ is nonzero, then AB and BA are similar.

$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are not similar.

7) Which of the following options are correct? ***1 point***

- [ ] Two invertible matrices of same order are equivalent.
- [ ] $A = \begin{bmatrix} 1 & 7 & 6 \\ 2 & 1 & -1 \\ 5 & 2 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -2 \\ 1 & 1 & 4 \\ 1 & 2 & -6 \end{bmatrix}$ are equivalent.
- [ ] $A = [\begin{smallmatrix} 2 & 3 \\ 4 & 6 \end{smallmatrix}]$ and $B = [\begin{smallmatrix} 1 & 3 \\ 3 & 1 \end{smallmatrix}]$ are equivalent.
- [ ] Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$ and $T_2(x, y, z) = (x + y, x - z, y + z)$. Let A and B denote the matrix representation of T_1 and T_2 with respect to the standard basis of R^3. Then A and B are equivalent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Two invertible matrices of same order are equivalent.

$A = \begin{bmatrix} 1 & 7 & 6 \\ 2 & 1 & -1 \\ 5 & 2 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -2 \\ 1 & 1 & 4 \\ 1 & 2 & -6 \end{bmatrix}$ are equivalent.

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$ and $T_2(x, y, z) = (x + y, x - z, y + z)$. Let A and B denote the matrix representation of T_1 and T_2 with respect to the standard basis of R^3. Then A and B are equivalent.

Level 2

8) Let E be the set of 2×2 matrices which are similar to a scalar matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$. **1 point**

Choose the set of correct statements.

- [ ] E is an infinite set.
- [ ] Cardinality of E is finite.
- [ ] E is singleton set.
- [ ] E consists of all the scalar matrices of order n.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality of E is finite.
E is singleton set.

9) Which of the following option(s) is(are) true? **1 point**

- [ ] If A and B are similar matrices then A^{-1} and B^{-1} are similar matrices.
- [ ] If A and B are similar matrices then A^2 and B^2 are similar matrices.
- [ ] If A^2 and B^2 are similar matrices then A and B are similar matrices.
- [ ] If A and B are similar matrices then A^T and B^T are similar matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A and B are similar matrices then A^{-1} and B^{-1} are similar matrices.
If A and B are similar matrices then A^2 and B^2 are similar matrices.
If A and B are similar matrices then A^T and B^T are similar matrices.

10) Let us consider the following matrices. **1 point**

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Choose the set of correct options.

- [ ] A and B have the same rank, and same determinant, but they are not similar matrices.
- [ ] A and C have the same rank, and same determinant, but they are not similar matrices.
- [ ] B and C have the same rank, and same determinant, but they are not similar matrices.
- [ ] None of the options are true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A and B have the same rank, and same determinant, but they are not similar matrices.
B and C have the same rank, and same determinant, but they are not similar matrices.

Your score is: 0/10